

Part 2: Evaluate a Genre Retrieval System

Luke Vrouwenvelder

To further improve song recommendations, the new approach implemented involved leaning into IDF by squaring the IDF values (IDF^2).

In practice, squared-IDF landed slightly below plain (un-weighted) cosine, and well below the original TF-IDF performance baseline. Squaring IDF values likely over-amplified a small selection of the rarest words, which resulted in matches that heavily favored a small overlap of highly weighted words, as opposed to a more even comparison of non-IDF words.

Because the IDF^2 values were multiplied directly into each song's score, squaring directly impacted the ranking profile for each song, and results for each song's recommendations were based on the rarest word in the song. This proved to be a low-quality recommendation device, even if, intuitively, it seemed viable.

Comparing the performance of both methods shows that the original TF-IDF method was a better balance of weighting and unweighting rare and common words, respectively, and a more robust method for recommendation.